

Limits Involving Infinity

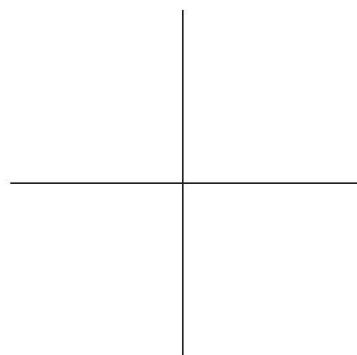
Section 1.6

A. Infinity versus DNE

Recall from Section 1.3 that $\lim_{x \rightarrow 0} \frac{1}{x^2}$ DNE since the function value kept increasing. Now we will be more descriptive; any value that keeps increasing is said to approach infinity (∞), and any value that keeps decreasing is said to approach negative infinity ($-\infty$).

Examples

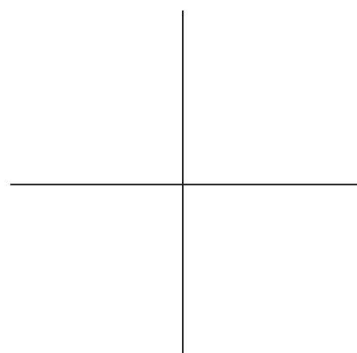
1. (video) Evaluate $\lim_{x \rightarrow 0} \frac{1}{x^2}$ using the graph and table method.



$\lim_{x \rightarrow 0^-} \frac{1}{x^2}$	
x	y
-0.1	
-0.01	
-0.001	

$\lim_{x \rightarrow 0^+} \frac{1}{x^2}$	
x	y
0.1	
0.01	
0.001	

2. (video) Evaluate $\lim_{x \rightarrow 0} \frac{1}{x}$ using the graph and table method.



$\lim_{x \rightarrow 0^-} \frac{1}{x}$	
x	y
-0.1	
-0.01	
-0.001	

$\lim_{x \rightarrow 0^+} \frac{1}{x}$	
x	y
0.1	
0.01	
0.001	

B. A Quick Review of Asymptotes

An asymptote is an imaginary line that the graph of a function approaches as the independent variable approaches a restricted number or infinity.

Locating Vertical Asymptotes

If $f(x) = \frac{p(x)}{q(x)}$ is a rational function, $p(x)$ and $q(x)$ have no common factors and k is a zero of $q(x)$, then the line $x = k$ is a vertical asymptote of the graph of $f(x)$.

Locating Horizontal Asymptotes

$$\text{Let } f(x) = \frac{p(x)}{q(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0}$$

- i. If $n < m$, then $y = 0$ is the horizontal asymptote.
- ii. If $n = m$, then the line $y = \frac{a_n}{b_m}$ is the horizontal asymptote.
- iii. If $n > m$, there is NO horizontal asymptote. (But there may be a slant/oblique asymptote.)

Examples: For the following rational functions, find the vertical and horizontal asymptotes if any:

3. $f(x) = \frac{16x^2}{4x^2 + 1}$

4. $g(x) = \frac{x+8}{x^2 - 64}$

5. $h(x) = \frac{x^3 + 7}{5x - 2}$

6. $k(x) = \frac{x^2 - 2x}{2 - 3x + x^2}$

C. Vertical Asymptotes

Vertical asymptotes occur when $\lim_{x \rightarrow a^-} f(x) = \pm\infty$ or $\lim_{x \rightarrow a^+} f(x) = \pm\infty$

The asymptote will be the line $x = a$.

Example: Evaluate the limit, find the asymptote and graph the function

7. (video) $\lim_{x \rightarrow 2} \frac{2x+1}{(3x-6)^2}$

8. (video) $\lim_{x \rightarrow -4^-} \frac{x+6}{x+4}$

$$\lim_{x \rightarrow -4^+} \frac{x+6}{x+4}$$

$$\lim_{x \rightarrow -4} \frac{x+6}{x+4}$$

9. $\lim_{x \rightarrow 2} \frac{x+1}{2x-4}$

D. Limits to Infinity

A limit as the domain approaches infinity: $\lim_{x \rightarrow \infty} f(x)$

Finding Limits to Infinity of Rational Functions

- i. Determine the degree of the denominator. (Let's say degree = P)
- ii. Multiply both the numerator and denominator by $\frac{1}{x^P}$.
- iii. Distribute/clean up algebra and continue evaluating the limit.

Example: Evaluate the limit.

10. (video) $\lim_{x \rightarrow \infty} \frac{6x^2 + 2x + 7}{8x + 2x^2}$

11. (video) $\lim_{x \rightarrow \infty} \frac{x^3 + 4x - 2}{6 - 2x^2}$

12. (video) $\lim_{x \rightarrow \infty} \frac{2x}{2x^2 + x - 1}$

Conclusion: For positive integers M and N such that $M > N$

1. Degree of the Numerator = Degree of the Denominator

$$\lim_{x \rightarrow \infty} \frac{\text{Polynomial of Degree } M}{\text{Polynomial of Degree } M} = \text{Ratio of Leading Coefficients}$$

2. Degree of the Numerator > Degree of the Denominator

$$\lim_{x \rightarrow \infty} \frac{\text{Polynomial of Degree } M}{\text{Polynomial of Degree } N} = \pm \infty$$

3. Degree of the Numerator < Degree of the Denominator

$$\lim_{x \rightarrow \infty} \frac{\text{Polynomial of Degree } N}{\text{Polynomial of Degree } M} = 0$$

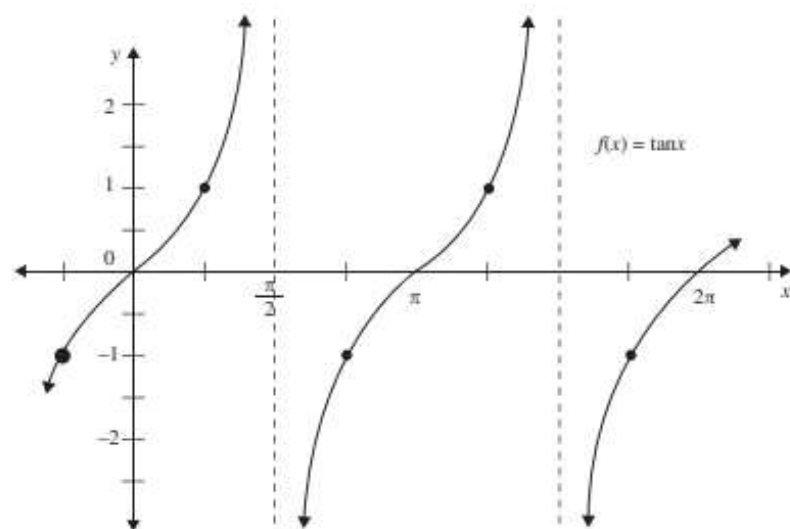
More Examples: Evaluate the limit.

13. $\lim_{x \rightarrow \infty} \frac{3x - 10}{\sqrt{16x^2 + 5}}$

14. $\lim_{x \rightarrow -\infty} \frac{\sqrt{9x^2 + x + 1}}{2x + 1}$

15. $\lim_{x \rightarrow \frac{\pi}{2}^-} \tan x$

16. $\lim_{x \rightarrow \frac{\pi}{2}^+} \tan x$



17. $\lim_{x \rightarrow \infty} \sqrt{x^2 + 7x + 1} - x$